



15.1

Arithmetic Mean (1)

15.1.1 Arithmetic Mean: ISRO CSE 2011 | Question: 67



The arithmetic mean of attendance of 49 students of class A is 40 and that of 53 students of class B is 35. Then the percentage of arithmetic mean of attendance of class A and B is

- A. 27.2% B. 50.25% C. 51.13% D. 37.4%

isro2011 probability arithmetic-mean

Answer key

15.2

Combinatory (1)

15.2.1 Combinatory: ISRO-DEC2017-3



A bag contains 19 red balls and 19 black balls. Two balls are removed at a time repeatedly and discarded if they are of the same colour, but if they are different, black ball is discarded and red ball is returned to the bag. The probability that this process will terminate with one red ball is

- A. 1 B. $\frac{1}{21}$ C. 0 D. 0.5

isrodec2017 probability combinatory

Answer key

15.3

Conditional Probability (3)

15.3.1 Conditional Probability: GATE CSE 1994 | Question: 1.4, ISRO2017-2



Let A and B be any two arbitrary events, then, which one of the following is TRUE?

- A. $P(A \cap B) = P(A)P(B)$ B. $P(A \cup B) = P(A) + P(B)$
 C. $P(A | B) = P(A \cap B)P(B)$ D. $P(A \cup B) \leq P(A) + P(B)$

gate1994 probability conditional-probability normal isro2017

Answer key

15.3.2 Conditional Probability: ISRO CSE 2007 | Question: 33



Company X shipped 5 computer chips, 1 of which was defective. and company Y shipped 4 computer chips, 2 of which were defective. One computer chip is to be chosen uniformly at a random from the 9 chips shipped by the companies. If the chosen chip is found to be defective, what is the probability that the chip came from the company Y?

- A. $\frac{2}{9}$ B. $\frac{4}{9}$ C. $\frac{2}{3}$ D. $\frac{1}{2}$

isro2007 probability conditional-probability

Answer key

15.3.3 Conditional Probability: ISRO-2013-71



Let $P(E)$ denote the probability of the occurrence of event E . If $P(A) = 0.5$ and $P(B) = 1$ then the values of $P(A|B)$ and $P(B|A)$ respectively are

A. 0.5, 0.25

B. 0.25, 0.5

C. 0.5, 1

D. 1, 0.5

isro2013 probability conditional-probability easy

Answer key

15.4

Mean Mode Median (1)

15.4.1 Mean Mode Median: ISRO CSE 2014 | Question: 74



What is the median of data if its mode is 15 and the mean is 30?

A. 30

B. 25

C. 22.5

D. 27.5

probability statistics mean-mode-median isro2014

Answer key

15.5

Normal Distribution (1)

15.5.1 Normal Distribution: ISRO CSE 2009 | Question: 65



If the mean of a normal frequency distribution of 1000 items is 25 and its standard deviation is 2.5, then its maximum ordinate is

A. $\frac{1000}{\sqrt{2\pi}} e^{-25}$ B. $\frac{1000}{\sqrt{2\pi}}$ C. $\frac{1000}{\sqrt{2\pi}} e^{-2.5}$ D. $\frac{400}{\sqrt{2\pi}}$

isro2009 statistics normal-distribution

Answer key

15.6

Poisson Distribution (1)

15.6.1 Poisson Distribution: ISRO CSE 2009 | Question: 66



If the pdf of a Poisson distribution is given by $f(x) = \frac{e^{-2} 2^x}{x!}$ then its mean is

A. 2^x

B. 2

C. -2

D. 1

isro2009 probability poisson-distribution

Answer key

15.7

Probability (5)

15.7.1 Probability: ISRO 2016 Mech: Probability



A and B are two candidates appearing for an interview by a company. The probability that A is selected is 0.5 and the probability that both A and B are selected is at most 0.3. The probability of B getting selected is

(a) 0.9

(b) ≤ 0.6 (c) ≤ 0.3

(d) 0.5

probability isro2016 engineering-mathematics

Answer key 

15.7.2 Probability: ISRO CSE 2011 | Question: 61



Three coins are tossed simultaneously. The probability that they will fall two heads and one tail is

- A. $\frac{5}{8}$ B. $\frac{1}{8}$ C. $\frac{2}{3}$ D. $\frac{3}{8}$

isro2011 probability

Answer key 

15.7.3 Probability: ISRO CSE 2014 | Question: 37



The probability that two friends are born in the same month is ____ ?

- A. $\frac{1}{6}$ B. $\frac{1}{12}$ C. $\frac{1}{144}$ D. $\frac{1}{24}$

probability isro2014

Answer key 

15.7.4 Probability: ISRO CSE 2018 | Question: 14



A class of 30 students occupy a classroom containing 5 rows of seats, with 8 seats in each row. If the students seat themselves at random, the probability that sixth seat in the fifth row will be empty is:

- a. $\frac{1}{5}$ b. $\frac{1}{3}$ c. $\frac{1}{4}$ d. $\frac{2}{5}$

isro2018 probability

Answer key 

15.7.5 Probability: ISRO CSE 2023 | Question: 53



If a bag contains 6 blue balls, 5 white balls and 4 yellow balls and if a ball is drawn out at random, probability of not getting a white ball is

- A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{1}{5}$ D. None of the above

isro-cse-2023 probability

Answer key 

15.8

Probability Density Function (1)

15.8.1 Probability Density Function: GATE CSE 2003 | Question: 60, ISRO2007-45



A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

- A. $f_1(t) + f_2(t)$ B. $\int_0^t f_1(x)f_2(x)dx$
C. $\int_0^t f_1(x)f_2(t-x)dx$ D. $\max\{f_1(t), f_2(t)\}$

gatecse-2003 probability normal isro2007 probability-density-function

Answer key 

15.9

Random Variable (1)

15.9.1 Random Variable: GATE CSE 2005 | Question: 12, ISRO2009-64



Let $f(x)$ be the continuous probability density function of a random variable x , the probability that $a < x \leq b$, is :

- A. $f(b-a)$
 B. $f(b) - f(a)$
 C. $\int_a^b f(x)dx$
 D. $\int_a^b xf(x)dx$

gatecse-2005 probability random-variable easy isro2009

Answer key

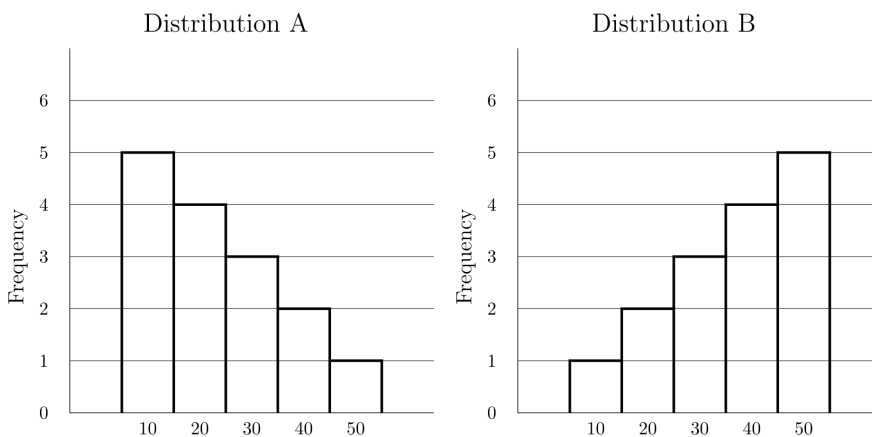
15.10

Standard Deviation (1)

15.10.1 Standard Deviation: ISRO CSE 2020 | Question: 56



For the distributions given below:



Which of the following is correct for the above distributions?

- A. Standard deviation of A is significantly lower than standard deviation of B
 B. Standard deviation of A is slightly lower than standard deviation of B
 C. Standard deviation of A is same as standard deviation of B
 D. Standard deviation of A is significantly higher than standard deviation of B

isro-2020 probability standard-deviation normal

Answer key

Answer Keys

15.1.1	D	15.2.1	A	15.3.1	D	15.3.2	C	15.3.3	C
15.4.1	B	15.5.1	D	15.6.1	B	15.7.1	B	15.7.2	D
15.7.3	B	15.7.4	C	15.7.5	Q-Q	15.8.1	C	15.9.1	C
15.10.1	C								